Simulation and Analysis of DAG-Based Distributed Ledgers

Tangle-Style Tip Selection and MCMC Stability Analysis

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Outline

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What is a Blockchain?

A **blockchain** is a distributed, append-only ledger in which transactions are grouped into blocks chained together by cryptographic hashes.

Key properties:

- Decentralization — no single controlling authority
- Immutability — confirmed data cannot be altered
- Consensus — PoW / PoS to agree on one chain
- Transparency — every participant audits the full history

Main applications:

- Cryptocurrencies (Bitcoin, Ethereum)
- Supply-chain tracking & logistics
- Smart contracts & DeFi
- Sharing data in a trustworthy manner
- IoT device orchestration

Problems with Blockchains

Performance

- **Low throughput:** Bitcoin $\approx$ 7 tx/s, Ethereum $\approx$ 15 tx/s. Visa averages $\approx$ 1,700 tx/s.
- **High latency:** Bitcoin block time $\approx$ 10 min; finality can take an hour.
- **Scalability ceiling:** sequential block production limits concurrency fundamentally.

Economics & Sustainability

- **Energy:** PoW mining rivals the electricity usage of small countries.
- **Fees:** miners demand fees, making micro-transactions uneconomical.
- **Centralization risk:** mining pools concentrate hash-power; economies of scale undermine decentralization.

These limitations motivate the search for alternative ledger architectures.

Introducing BlockDAGs

A **Directed Acyclic Graph (DAG)**-based ledger replaces the single chain: each new transaction can reference *multiple* previous ones simultaneously.

Advantages over blockchains:

- **Parallel processing** — many transactions issued concurrently, no block-slot bottleneck.
- **Scalable throughput** — throughput grows with the number of active participants.
- **No dedicated miners** — every transaction validates previous ones (self-serving consensus).
- **Fee-free potential** — PoW is lightweight and per-transaction; no block reward necessary.

Particularly attractive for **IoT**, smart-city applications, and high-frequency micro-trading where low fees and low latency are essential.

BlockDAGs vs. Blockchains

	Blockchain	**BlockDAG**
Structure	Linear chain	Directed acyclic graph
Concurrency	1 block per slot	Many txs in parallel
Throughput	$\sim$7–15 tx/s	Scales with users
Micro-txs	Impractical (fees)	Near-zero cost
Confirmation	Minutes	Seconds
IoT suitability	Heavy PoW, high fees	Lightweight PoW, fee-free

BlockDAGs: Limitations

BlockDAGs are *not* a complete replacement for blockchains:

- **Smart-contract complexity:** state-dependent execution needs a total order on transactions; DAGs make global ordering harder.
- **Probabilistic finality:** most DAG protocols provide weaker finality guarantees than PoS chains — problematic for financial settlement.
- **Ecosystem maturity:** tooling, auditing frameworks, and developer libraries lag far behind Ethereum / Solidity.
- **Coordinator dependency:** IOTA historically relied on a centralized Coordinator; fully decentralized DAG consensus remains an open research problem.
- **Double-spend vulnerability:** without a carefully designed tip-selection rule, attackers can exploit the DAG structure. The tension between security parameter α and liveness is non-trivial — and is precisely what motivates this project.

The Tangle Architecture

The **Tangle** [1] is IOTA's DAG-based DLT, designed for feeless IoT micro-transactions.

- Each new transaction ("site") selects and approves $m = 2$ previous transactions.
- Unapproved transactions form the **tip set** — the growing frontier of the DAG.
- Before being added, a transaction must complete a lightweight **Proof of Work** taking time h.
- The **cumulative weight** of a site equals the number of transactions that approve it (directly or indirectly).

Central metric: the tip count $L(t)$ at time t.

- $L(t)$ bounded $\Rightarrow$ all transactions are eventually approved.
- $L(t) \to \infty \Rightarrow$ orphaned transactions accumulate.

Tip Selection Algorithms

Random Selection (RS)

- Pick m tips uniformly at random.

+ Simple; $L(t)$ stays bounded.

− Vulnerable to double-spend attacks.

The core dilemma:
Security ($\alpha\uparrow$) vs. Liveness ($\alpha\downarrow$).

MCMC Selection

- Start m random walks at the genesis.
- At site j, jump to child k with probability

$$P(j \to k) \propto e^{-\alpha(\vartheta_j - \vartheta_k)}$$

  where ϑ_i is the cumulative weight.
- Walk terminates at a tip.

+ Resistant to double-spend.

− High $\alpha \Rightarrow$ old tips orphaned.

MCMC Walk — Step by Step

1. **Initialise:** launch $m = 2$ independent walkers at the genesis site.
2. **Transition:** at site j with children $\{k_1, \ldots, k_d\}$, move to k_i with probability

$$P(j \to k_i) = \frac{\exp\left(-\alpha\left(\vartheta_j - \vartheta_{k_i}\right)\right)}{\sum_{k \in \text{children}(j)} \exp\left(-\alpha\left(\vartheta_j - \vartheta_k\right)\right)}$$

3. **Terminate:** when the walker reaches a tip (a site with no children), select that tip for approval.
4. **Attach:** new transaction creates edges to its m selected tips and performs PoW (delay h).

Effect of α:

- $\alpha \to 0$: transition probabilities become uniform $\Rightarrow$ equivalent to random selection.
- $\alpha \to \infty$: walker follows the single heaviest-weight path $\Rightarrow$ deterministic, greedy.

Fluid Model — Setup

Ferraro et al. [2] derive a **mean-field PDE** for the Tangle in the fluid limit ($\lambda \to \infty$, Poisson arrivals at rate λ).

Key quantities (rescaled by λ^{-1}):

$$l(t) = \lim_{\lambda\to\infty} \lambda^{-1} L(t), \qquad x(t) = \lim_{\lambda\to\infty} \lambda^{-1} X(t), \qquad w(t) = \lim_{\lambda\to\infty} \lambda^{-1} W(t)$$

where $L(t)$ = tips, $X(t)$ = free tips, $W(t)$ = pending tips ($W \leq mh$).

Age-density $l(t, s)$: number of tips at time t with age s (time since added to DAG).

Selection model: each selection j uses weight function $g_j(s)$ with normalisation

$$\zeta_j(t) = \int_0^{t-h} g_j(s)\, l(t, s)\, \mathrm{d}s$$

- Random: $g(s) = 1$ (constant, *not* integrable).
- MCMC-like: $g(s)$ decays with age (*integrable* for large α).

Fluid Model — The Delayed PDE System

Evolution of free-tip density ($s > 0$):

$$\boxed{\frac{\partial x}{\partial t}(t,s) + \frac{\partial x}{\partial s}(t,s) \;=\; -\,x(t,s)\sum_{j=1}^{m}\frac{g_j(s)}{\zeta_j(t)}}$$

Evolution of total tip density:

$$\boxed{\frac{\partial l}{\partial t}(t,s) + \frac{\partial l}{\partial s}(t,s) \;=\; -\,x(t-h,\; s-h)\sum_{j=1}^{m}\frac{g_j(s-h)}{\zeta_j(t-h)}\;\cdot\;\theta(s-h)}$$

with boundary conditions $x(t,0) = l(t,0) = 1$ and

$$\zeta_j(t) = \int_0^{t-h} g_j(s)\, l(t,s)\,\mathrm{d}s.$$

These are **delayed PDEs**: the density at time t depends on the state at time $t-h$ (the PoW delay).

Fluid Model — Key Theorems

Theorem 2 (Integrable weights $\Rightarrow$ orphan persistence):
If all g_j satisfy $\int_0^\infty g_j(s)\,\mathrm{d}s < \infty$, then any steady-state solution satisfies

$$\int_0^\infty I(s)\,\mathrm{d}s \;=\; \infty.$$

Interpretation: under pure MCMC with large α, the weight $g(s)$ decays fast (integrable), so selection concentrates on recent tips. Old tips are never picked $\Rightarrow$ orphan count diverges.

Theorem 3 (Random selection $\Rightarrow$ bounded tips):
If at least two selections are uniform random ($g_1 = g_2 = 1$), then

$$I(t) \;\leq\; C \quad \text{for all } t \geq 0.$$

All transactions are eventually approved. However, random selection is vulnerable to double-spend attacks.

The Hybrid Tip-Selection Algorithm

Ferraro et al. [2] propose a **hybrid** algorithm that resolves the security–liveness tension:

1. **Security step:** first selection uses MCMC with a *high* α.
   - Ensures that dishonest tips on low-weight paths are very unlikely to be chosen.
   - Protects against double-spend attacks.
2. **Swipe step:** second selection uses random selection (or MCMC with a very *low* $\alpha \approx 0$).
   - Guarantees that old, forgotten tips are eventually picked.
   - Keeps $L(t)$ bounded $\Rightarrow$ no orphans.

Result: the hybrid algorithm inherits the security of high-α MCMC *and* the stability guarantee of random selection.

Project Description

We implemented and experimentally validated the Tangle tip-selection model from Ferraro et al. [2] with **two independent systems**:

`tangle-sim` (benchmark)

- Single-process Python/asyncio simulator.
- Discrete-event engine; priority-queue scheduling.
- 66 runs, 13 074 transactions.
- Campaign time: ∼35 min.

`tangle-distributed` (validation)

- One OS process per node; TCP sockets.
- Real gossip protocol over the network.
- 47 runs, 9 224 transactions.
- Campaign time: ∼25 min.

Total: 113 runs | 22 298 transactions | ∼55 min wall-clock.

How We Improved Upon the Paper

Ferraro et al. used a single-process agent-based simulation. We extended the work in several directions:

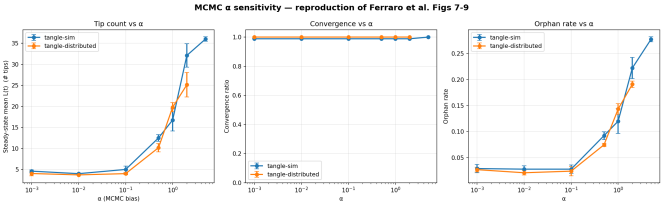
- **True distributed implementation:** one OS process per node with TCP-socket gossip — real network I/O, not simulated.
- **Multiple topologies:** full-mesh, ring, small-world, random-k, star (paper assumed full connectivity).
- **Realistic network model:** log-normal latency with configurable base (20–800 ms) and jitter.
- **Adversarial testing:** one node issuing at $4\times$ the honest rate; the paper treats double-spend only theoretically.
- **Cross-validation:** two independent codebases producing statistically identical results — strong evidence of correctness.
- **Broader sweeps:** seven parameter dimensions (α, algorithm, λ, h, n, topology, latency) totalling 113 runs versus the paper's focused simulations.

Experiment Design

Campaign	Parameter varied	Runs (sim / dist)
MCMC α sensitivity	$\alpha \in \{0.001,\ 0.01,\ 0.1,\ 0.5,\ 1,\ 2,\ 5\}$	14 / 12
Algorithm comparison	Random, MCMC-low, MCMC-high, Hybrid	8 / 8
Arrival rate λ	$\lambda \in \{0.5,\ 1,\ 2,\ 5,\ 10\}$ tx/s per node	10 / —
PoW delay h	$h \in \{0.1,\ 0.3,\ 0.6,\ 1.0\}$ s	8 / —
Node count n	$n \in \{3,\ 5,\ 8,\ 12\}$	8 / 6
Topology	full-mesh / ring / small-world / random-k / star	10 / 10
Latency	base $\in \{20,\ 100,\ 300,\ 800\}$ ms	8 / 8
Attack	1 adversary at 4λ, 6 honest	— / 2

Baseline: $n=5$, $\lambda=2$ tx/s, $h=0.2$ s, $m=2$, full-mesh, log-normal latency (base 100 ms), $T=25$ s, Hybrid.

Result: MCMC α Sensitivity

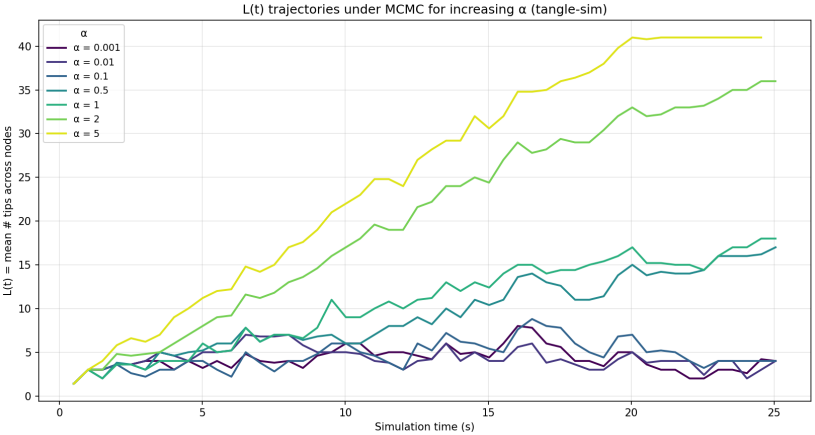


Findings:

- Tip count explodes as α crosses the $0.1 \to 1$ region.
- $\alpha = 0.001$: $\bar{L} \approx 4.6$ / 4.0
- $\alpha = 1.0$: $\bar{L} \approx 16.8$ / 19.8
- $\alpha = 5.0$: $\bar{L} \approx 36$ (unstable)
- Convergence remains >99% even at high α.

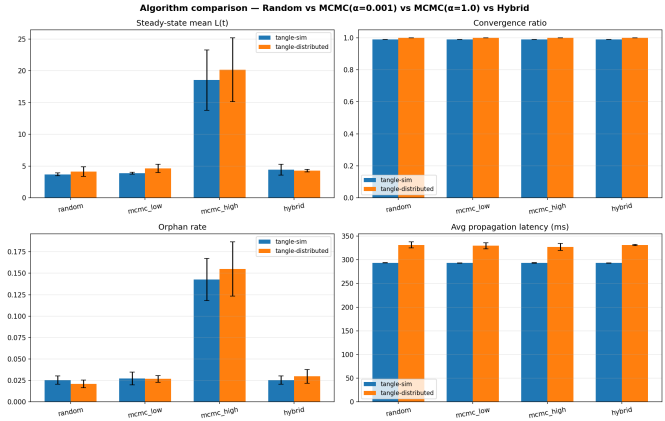
Reproduces Ferraro Figs. 7–9.

Result: $L(t)$ Trajectories by α



Low α: $L(t)$ stabilises near $2\lambda h$. High α: $L(t)$ diverges.

Result: Algorithm Comparison

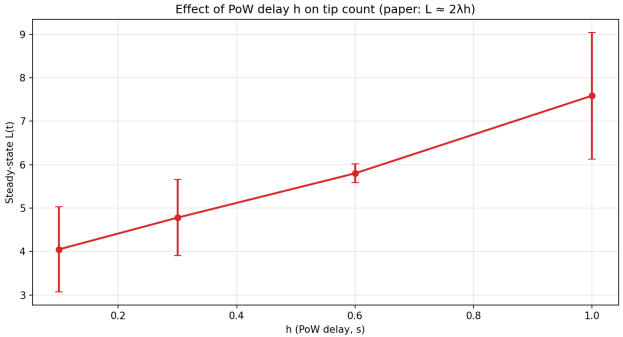


Mean $\bar{L}$ (sim / dist):

- Random: 3.69 / 3.81
- MCMC-low: 3.89 / 4.02
- MCMC-high: 18.54 / 19.80
- **Hybrid:** 4.44 / 4.20

Hybrid stays near Random's $\bar{L}$ while using $\alpha = 1$ for security — **core claim confirmed.**

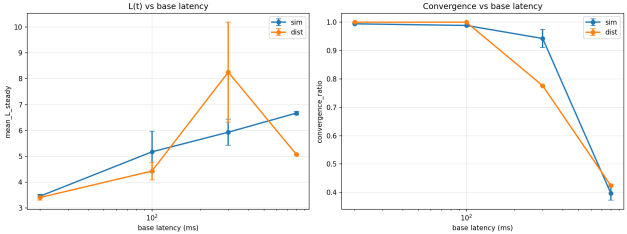
Result: PoW Delay h



$\bar{L}$ vs. PoW delay h. 8 sim runs.

- $\bar{L}$ grows roughly linearly with h, as the paper predicts.
- At $h = 0.1$ with seed 42, convergence dipped to 75%: tip-selection races become tight when PoW is fast relative to propagation.

Result: Latency Sensitivity



$\bar{L}$ and convergence vs. base latency.

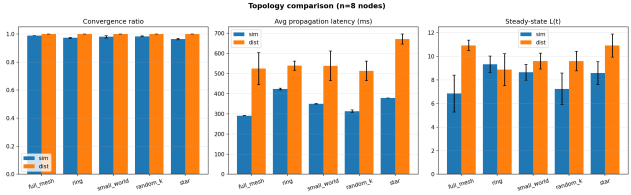
Most sensitive parameter:

- 20 ms: conv. 100%, $\bar{L} \approx 3.5$
- 100 ms: conv. 99%, $\bar{L} \approx 4.4$
- 300 ms: conv. drops to **77%**
- 800 ms: conv. drops to **42%**

When network latency approaches h, liveness breaks.

The paper does not analyse this dimension — this is a new operational finding.

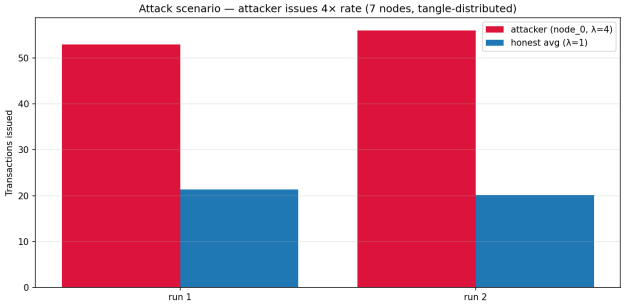
Result: Topology



Topology comparison at $n = 8$.

- All five topologies converge in the distributed setting.
- Full-mesh propagates fastest $\Rightarrow$ smallest tip window.
- Ring and star are worst (gossip takes $O(n)$ hops).
- Topology affects speed, **not correctness**.

Result: Attack Resistance



Adversary vs. honest transaction counts.

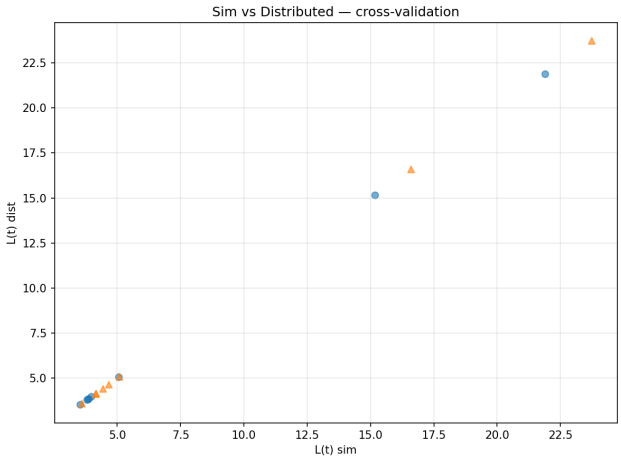
Setup: 7 nodes; 1 adversary at 4× honest rate.

Outcome:

- Adversary issued ∼54 txs vs. ∼21 per honest peer.
- **100% convergence** in both runs.
- Honest nodes picked adversary tips at ∼50% the naive rate.

The hybrid selector's MCMC step naturally deprioritises adversarial tips.

Cross-Validation: Sim vs. Distributed



Scatter of per-run metrics across both implementations.

Aggregate statistics:

- Mean convergence: sim 96.2%, dist 96.6%.
- Mean $\bar{L}$: sim 8.41, dist 8.35.
- Difference: **0.7%**.

The asyncio simulator is a faithful proxy for the true multi-process system.

Long run (60 s): 416 txs, $\bar{L} = 5.04$, 100% convergence — matches 25 s baselines.

Validating the Paper's Claims

1. **High-α MCMC $\Rightarrow$ orphan persistence.**
   ✓ Confirmed. Tip count grows $\sim 8\times$ from $\alpha = 0.001$ to $\alpha = 2$ in both implementations.

2. **Hybrid keeps $L(t)$ bounded while preserving security.**
   ✓ Confirmed. Hybrid $\bar{L} = 4.3$ vs. MCMC-high $\bar{L} = 19.2$; security step uses $\alpha = 1$.

3. **Mean-field prediction $L \approx 2\lambda h$.**
   ✓ Confirmed. $\bar{L}$ scales linearly with λ and h across all configurations.

4. **System is stable at equilibrium.**
   ✓ Confirmed. 60 s run matches 25 s baselines exactly.

5. **New finding: Network latency is the critical operational parameter.**
   Convergence drops to 42% at 800 ms. Not analysed in the original paper.

Conclusions and Future Work

Summary:

- Built two independent Tangle implementations (asyncio simulator + distributed TCP system).
- Reproduced all key predictions of Ferraro et al. across 113 runs and 22 298 transactions.
- Confirmed the hybrid tip selector delivers bounded $L(t)$ with high-α security.
- Identified network latency as the critical real-world sensitivity parameter.

Future directions:

- Multi-attacker and coordinated attack scenarios.
- Confirmation-time and path-length metrics from genesis.
- Replace full broadcast with gossip / partial-broadcast protocols.
- Move from simulation to deployment on a real network testbed.

References

S. Popov, "The Tangle," *IOTA Foundation White Paper*, version 1.4.3, April 2018.

P. Ferraro, C. King, and R. Shorten, "On the stability of unverified transactions in a DAG-based Distributed Ledger," *IEEE Transactions on Automatic Control*, vol. 65, no. 9, pp. 3772–3783, September 2020. `https://arxiv.org/abs/1901.07302`